RAMADHINA NURDIANTI

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Senior GIS Analyst with a focus on agricultural sustainability and over five years of practical experience in GIS, remote sensing, and data visualisation. Proficient in ArcGIS Pro and Python, with developing expertise in front-end Web GIS development. Experienced in creating high-quality maps and data visualisations from extensive and complex datasets. Skilled in conducting geospatial analyses and applying statistical methods to derive meaningful insights. Committed to leveraging geospatial technology to support environmental conservation and sustainability, currently contributing to global sustainability initiatives at Earthworm Foundation.

WORK EXPERIENCES

Earthworm Foundation – Pichincha, Ecuador **Jan 2026 – *present***

Senior GIS Analyst – Global Data Team

- Develop spatial methodologies to identify areas suitable for commodity expansion and potential forest-risk areas, supporting early identification of risks to remaining forests.
- Develop map design guidelines to improve consistency, clarity, functionality, and visual quality across organisational maps and spatial products.
- Use Python to automate repetitive GIS and data-processing workflows, improving efficiency and reducing manual processing.
- Coordinate a four-person GIS team responsible for quarterly deforestation monitoring reports across six organisational landscapes, overseeing project planning, task coordination, quality review, and delivery of spatial outputs.
- Serve as the focal point for content and visual alignment for the quarterly deforestation reporting project, developing map templates and automated reporting workflows, including Excel/VBA tools to transfer charts into PowerPoint templates.
- Develop interactive Web GIS maps for organisational landscapes in Brazil, improving access to spatial information and supporting field teams with field-visit planning and preparation.

Earthworm Foundation – Semarang, Indonesia **Jan 2023 – Dec 2025**

Senior GIS Analyst – Global Data Team

- Investigated commodity supply chains and traceability for oil palm, cocoa, and pulp & paper, combining public and supplier-provided data to assess sourcing origins, supply-chain relationships, and sustainability risks.
- Conducted deforestation alert validation across member supply chains and organisational landscapes, assessing potential impacts on peatlands, HCV, and HCS areas across oil palm, cocoa, and pulp & paper concessions.
- Collected and managed data on concessions, sourcing areas, processing facilities, and conservation areas to support supply-chain and landscape-level investigations.
- Supported grievance investigations through spatial analysis and validation of reported concerns related to members' supply chains and potential environmental impacts.
- Conducted GIS and remote sensing analyses for client projects and internal assessments, including land-cover change, sourcing-area risk, forest footprint, and landscape-level environmental analysis.

Forestry Education and Training Unit of Sebelas Maret University - Karanganyar, Indonesia **Feb 2022 – Jul 2022**

Mapping and Data Processing Intern

- Mapped forest areas using the DJI Mavic Pro Platinum drone as well as create orthomosaic and DEM data using Agisoft Metashape and PCI Geomatica.
- Identified and managed data of land boundaries, plant types, and the identity of forest farmers from drone aerial imagery and field data.
- Maintained forest area boundary markers and reported on the condition of the most recent border markers.
- Mapped trail motorbike tracks in the forest areas using Garmin GPS.
- Organized and conducted forestry advising for forest farmer groups.
- Coordinated events and activities at KHDTK Gunung Bromo UNS.

PT. Perkebunan Nusantara VIII (Kertajaya Oil Palm Plantation) - Lebak, Indonesia **Jan 2021 – Feb 2021**

GIS Analyst Intern

- Updated the maps and data for planting block areas.
- Estimated the age and condition of oil palm crops using the NDVI algorithm using Landsat 8 imagery.
- Assisted with the coordination of harvesting and plant management activities.

Sebelas Maret University - Surakarta, Indonesia **Feb 2021 – Jul 2022**

Course Assistant Remote Sensing & Geographic Information System

- Mentored on how to classify land use in Landsat 8 imagery.
- Mentored on how to operate DJI Phantom 4 drone, process, and analyze aerial photos.
- Mentored on how to georeference, digitize, layout maps, and build Web GIS applicationsusing ArcGIS Online.

PUBLICATIONS AND SKILLS

Publications

- Agricultural Land Dryness Distribution Using the Normalized Difference Drought Index (NDDI) Algorithm on Landsat 8 Imagery in Eromoko, Indonesia – ENRJ (2023) – https://ph02.tci-thaijo.org/index.php/ennrj/article/view/248366

Projects

- Place-Based Initiative in Ecuador and Ghana: Analyse potential district areas to be developed as focus areas for a sustainability initiative with cocoa farmers (September 2026) – Spatial analysis and visualisation for an international FMCG company from the USA – (private document)
- Sourcing Area Deforestation Risk in Five Cocoa-Producing Countries: Analyse the potential sourcing areas of cocoa collection centres and their deforestation risks associated with commodity farming expansion (October & December 2025) – Spatial analysis and visualisation for an international FMCG company from the USA – (private document)
- Peat and Forest Assessment in Five Palm Oil-Producing Countries: Provide an overview of peatlands and forests exposed to supply-chain deforestation risks due to future commodity expansion (December 2024) – Spatial analysis and visualisation for a food and beverage manufacturing company based in Colombia – (private document)
- Sunflower Production in Uganda: Provide an overview of sensitive ecosystems and supply-chain exposure to deforestation (May 2024) – Spatial analysis, visualisation and report writing for an edible fats producer based in Switzerland – (private document)
- Palm Oil Forest Footprint in Riau Province: Rokan Hilir, Bengkalis, Dumai (April 2024) – Spatial analysis and visualisation report for an international FMCG company from Japan – (link to the document)
- Mangrove Distribution and Density: Padang City, Indonesia (September 2021) – Remote sensing analysis and map visualisation for a private-sector company in Indonesia – (link to webapp)

Language Skills

- Indonesian (Native)
- English (C1/ Advanced) (The official EF SET Certificate™ | Ramadhina Nurdianti)
- Spanish (A2/ Basic Conversation)

Courses and Training

- Advanced Drones Mapping (Advanced 1st Level) May 2022
- Basic Drones Mapping Mar 2022

Technical Skills

- **GIS & Spatial Analysis:** ArcGIS Pro, ArcGIS Online, QGIS, spatial analysis, geoprocessing, spatial data management.
- **Programming & Automation:** Python, ArcPy, SQL, Excel VBA.
- **Front-End Web GIS Development:** HTML, CSS, JavaScript.
- **Data Analysis & Visualisation:** Microsoft Excel, Power BI, IBM SPSS Statistics.
- **Photogrammetry & Image Processing:** Agisoft Metashape, PCI Geomatica.
- **Field Data Collection:** Garmin GPS, drone data collection.

TEACHING EXPERIENCES

Kampus Merdeka GIS Course Mentor - Surakarta, Indonesia **Sep 2022 – Oct 2022**

- Demonstrated how to set up and check that the drones (DJI Mavic Pro and DJI Mavic Air 2) are in accordance with the guidelines.
- Perform acquisition and quality control to ensure that the results of aerial photography meet the established standards and objectives.
- Processing aerial photo imagery to create orthomosaic and DEM data using Agisoft Metashape and PCI Geomatica.
- Mentored on how to georeferencing and create map layout.

Advanced Class GIS Training Speaker - Surakarta, Indonesia **Sep 2021**

- Mentored on how retrieving public spatial data set.
- Mentored on how to use the USLE method to create erosion risk maps based on spatial data analysis.

EDUCATION LEVEL

Sebelas Maret University - Surakarta, Indonesia **Aug 2018 – Jul 2022**

Undergraduate in Soil Science – Faculty of Agriculture

- GPA : 3.69/4.00